



1598 - Analysis of GTO Observations of Neptune by the MIRI Instrument

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Observations of Neptune by the MIRI instrument made in the GTO program will be analyzed for spatial variability of temperatures and composition. The scientific questions the program addresses emerge from previous and ongoing work using spacecraft and ground-based observations of Neptune's atmosphere. This work will (1) produce spatially resolved unambiguous maps of Neptune's stratospheric temperatures and composition, constraining the 3-dimensional structure and circulation of Neptune's middle atmosphere; (2) establish relationships between Neptune's visible structure and its 3-dimensional temperature and wind field, and (3) examine the relationship between storms and waves in the atmosphere. This work will help understand differences between dynamics of Neptune's and Saturn's atmosphere, both with a similar seasonal forcing arising from nearly the same axial tilt. The determination of Neptune's temperature structure will enable a refinement of far-infrared through submillimeter spectral models of Neptune, which serves as one of several primary calibrators for millimeter and submillimeter astronomy, e.g. for ALMA. The results are also important because Neptune is viewed as an archetype for similar-mass exoplanets known to exist in our galaxy from Kepler observations.

OBSERVING DESCRIPTION

The analysis will address archived observations by the MIRI experiment made in the GTO program 1249 for which Heidi Hammel is the PI and which has been implemented by Leigh Fletcher.

The observations consist of three sets of measurements by all 12 settings of the MIRI instrument using medium-resolution spectroscopy in IFU mode with four-point dithering to ensure coverage of the whole disk. Each set of measurements will be made at approximately 120 degrees of longitude separation in order to cover the planet at all longitudes during its 16h 14m rotation. On the advice of the MIRI team, one observation of the background sky will also be taken.

Program 1249 also includes NIRSPEC observations, which will not be addressed by this program except to supply context in reflected sunlight should a major atmospheric feature in the MIRI spectral images be detected.